



PHARMACEUTICAL MANUFACTURING COMPANY

A pharmaceutical manufacturing company faced challenges in maintaining data integrity and complying with FDA's 21 CFR Part 11. Ensuring secure, auditable electronic records when managing large volumes of lab samples proved difficult. Key issues included unauthorized access, outdated legacy systems, complex regulations, time-consuming validation, etc.

To address these issues, the company adopted **Logilab ELN**, cloud-based electronic lab notebook, specifically designed to digitize and facilitate laboratories to create an efficient, audit-ready environment.

Through a meticulous shift, the transition was managed with minimal operational disruption. As a result, the company achieved more streamlined, compliant lab operations, reinforced trust in its data practices, and regained the ability to meet global regulatory requirements effectively.



WHAT'S UP WITH PHARMACEUTICAL LABORATORIES

A pharmaceutical company involved in manufacturing active pharmaceutical ingredients (APIs), advanced intermediates, etc., faced significant challenges in complying with regulatory compliance requirements, particularly FDA's 21 CFR Part 11.

Major operational hurdles included,

Data Integrity & Security: Ensuring the integrity, authenticity, and confidentiality of lab data, and reducing the risk of unauthorized access or tampering.

Regulatory Complexity: Difficulty understanding and applying 21 CFR Part 11 requirements due to lack of comprehensive knowledge.

Legacy Systems: Existing infrastructure was outdated and lacked compliance support.

Documentation & Validation: Manual processes for system validation and audit trails were time-consuming and prone to errors.

Cost of Compliance: The financial burden of implementing and maintaining compliance measures was substantial. This included costs for system validation, employee training, system modifications, and third-party assessments to ensure ongoing compliance.

HOW LOGILAB ELN ENABLES SEAMLESS REGULATORY COMPLIANCE

Q: What if you had a system that facilitates your lab to be audit-ready and compliant to regulatory compliance requirements while streamlining lab operations?

A: Logilab ELN is a cloud-based Electronic Lab Notebook that provides comprehensive audit trails, version control, role-based access control, electronic signatures for approvals, etc., to enable your lab to be efficient, secure, and compliant.

GET THE PICTURE

The pharmaceutical company's laboratory processes were complex, handling a high volume of samples on a daily basis. As part of the highly regulated pharmaceutical industry, it was essential to maintain complete data integrity and ensure that all systems were fully compliant with 21 CFR Part 11 requirements. However, they found it difficult to comply due to risks around data security, regulatory complexity, poor infrastructure systems, and the burden of extensive documentation. Overcoming these challenges demanded robust digital infrastructure, process modernization, and specialized compliance knowledge.

AUTOMATED PHARMACEUTICAL LABORATORIES

To address these challenges, the pharmaceutical company adopted Agaram Technologies' **Logilab ELN**, a cloud-based electronic lab notebook solution. With capabilities such as configurable sheet and protocol templates, role-based access control, secure cloud storage, auto-versioning, comprehensive audit trails, and electronic signatures, Logilab ELN was the perfect solution to streamline lab operations and facilitate full compliance with 21 CFR Part 11.

To ensure effective system setup, the implementation team began by evaluating the legacy systems, aiming to upgrade them to a fully paperless, error-free environment that meets stringent FDA compliance standards.

ASSESSMENT & CONFIGURATIONS

The implementation team from Agaram Technologies conducted a thorough assessment of the pharmaceutical company's existing laboratory processes and systems to identify compliance gaps and opportunities to enhance efficiency and data integrity through digitalization. Based on this evaluation, a detailed implementation plan was developed.

Logilab ELN was then configured to meet the company's specific operational and regulatory requirements. This included setting up role-based access controls to restrict data access to authorized personnel, enabling electronic signatures in compliance with FDA guidelines, and implementing audit trails for complete traceability of all record modifications. Customizations and enhancements were completed to ensure the system aligned with the pharmaceutical company's processes and regulatory requirements.

EFFORTLESS DATA MIGRATION

Migrating existing data from legacy systems to Logilab ELN was a critical step. The implementation team ensured that all historical data was accurately transferred, validated, and integrated into the new system without compromising data integrity. This process included extensive testing to verify the accuracy and completeness of the migrated data.

TRAINING & SUPPORT

Comprehensive training programs were conducted to equip the pharmaceutical company's staff with the knowledge and skills to use Logilab ELN effectively. The training focused on the proper handling of electronic records, application of electronic signatures, and adherence to data integrity best practices. Ongoing support was also provided during the initial implementation phase to resolve any issues and ensure a smooth and successful transition to the new system.

VALIDATION & COMPLIANCE ASSURANCE

To ensure compliance with 21 CFR Part 11, extensive validation procedures were conducted. This included testing the system's functionality, verifying that it met all regulatory requirements, and ensuring that it could consistently produce reliable and accurate data. Regular audits and validations were scheduled to maintain ongoing compliance and address any potential issues promptly.

ONGOING MONITORING & MAINTENANCE

After the initial implementation, the system was continuously monitored to ensure it operated as intended. Feedback from users was collected to make necessary adjustments and improvements. Regular updates and maintenance were carried out to keep the system compliant with evolving regulatory requirements and technological advancements.

BUSINESS USE CASES

INWARD LOGBOOK

An inward log book in the pharmaceutical industry is a crucial document used to record and track the receipt of various materials, goods, and supplies entering the facility. This includes raw materials, packaging materials, and

other items necessary for production and research activities. With Logilab ELN's labsheets, customizable templates to record key data like receipt time, supplier info, batch details, inspection status, etc., can be created.

E-SAMPLE MANAGEMENT

Templates to record sample IDs, associated tests, order information, date/time of processing, etc., with all required validation and approval workflow processing can be created with Logilab ELN. By digitizing the entire sample lifecycle, it reduces manual errors, enhances traceability, and improves overall laboratory efficiency.

LAB WORK ALLOCATION

Streamline lab work allocation based on personnel skills and availability, ensuring efficient resource management, including equipment and supplies with Logilab ELN. It enables dynamic scheduling and prioritization of tasks such as allocation of tests to users, with real-time monitoring and automated notifications to keep track of progress.

QUALITY MANAGEMENT SYSTEM LOGS

Logilab ELN can digitalize quality management system logs by centralizing all quality-related data and documentation. It enables creation, management, and real-time tracking of quality logs, including deviations, CAPAs, and audit trails. Features like automated workflows, electronic signatures, and secure access controls support compliance with FDA 21 CFR Part 11.

CUSTOMIZED CALCULATIONS & CHEMICAL TESTING

Logilab ELN enables digitalization of customized calculations for various testing parameters and chemical testing by integrating calculation templates and automated data entry. It allows configuration of specific formulas tailored to unique testing protocols. Test data can be entered directly into the ELN, where calculations are performed automatically, ensuring accuracy and consistency. The system also supports real-time data validation and error checking, reducing manual errors.

COLUMN MANAGEMENT

Logilab ELN optimizes chromatography column management by facilitating the maintenance of digital records of each chromatography column, including details such as serial number, manufacturer, specifications, and validation status. It tracks usage, maintenance schedules, and performance data, with automated alerts for revalidation or replacement.

IMPROVED PROCESSES IN THE PHARMACEUTICAL LABORATORIES DUE TO IMPLEMENTATION OF LOGILAB ELN

- Auto-versioning, audit trails, and e-signatures ensure full compliance with regulations like FDA 21 CFR Part 11.
- Faster and more accurate sample management enabled by labsheet features.
- Easy and precise monitoring of inventory consumption, reorder levels, and expiry dates.
- Improved data access, data security, and collaboration due to Logilab ELN's cloud-based system.

KEY BENEFITS

- **Enhanced User Experience:** Better user experience and confidence due to simplified but feature-rich UIs.
- **Faster Implementation:** Save time in terms of implementation and roll-out due to cloud-based SaaS model.
- **Operational Efficiency:** Operational time savings and cost due to automated and paperless processes.
- **Cost-Effective Ownership:** Low cost of ownership and reduced overhead costs due to standard technology architecture.

- **Regulatory Compliance:** Low compliance cost due to adherence to regulatory and industry standards of data integrity.
- **Improved Accuracy:** Minimum errors due to automated processes and consistent result generation.

By implementing Logilab ELN, the pharmaceutical manufacturing company improved sample management, data integrity, and lab efficiency through a secure, scalable, and compliant cloud-based digital solution.

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